

US Patent & Trademark Office

Patent Public Search | Text View

United States Patent Application Publication

20250282599

Kind Code

A1

Publication Date

September 11, 2025

Inventor(s)

Ferranti; Armand Joseph et al.

CONTAINER OPENER

Abstract

A container opener includes a clamp that holds a container; a puncturing device that punctures a lid of the container held in the clamp; a blade that cuts the lid from the container while the blade and the container rotate with respect to each other; and a separating device that separates the puncturing device from the lid.

Inventors:	Ferranti; Armand Joseph (Long Branch, NJ), Caracappa; Alexander Armand (Medford, MA), Turner; Lyra (E. Montpelier, VT)
Applicant:	Draft Top, Inc. (Ewing, NJ)
Family ID:	1000008631456
Appl. No.:	18/860073
Filed (or PCT Filed):	April 26, 2023
PCT No.:	PCT/US2023/019944

Related U.S. Application Data

us-provisional-application US 63335518 20220427

Publication Classification

Int. Cl.: **B67B7/00** (20060101)

U.S. Cl.:

CPC **B67B7/38** (20130101);

Background/Summary

CROSS REFERENCE TO RELATED APPLICATIONS [0001] This application claims the benefit of U.S. Provisional Patent Application No. 63/335,518, filed Apr. 27, 2022, which is hereby incorporated by reference for all purposes as if fully set forth herein.

FIELD

[0002] The present disclosure generally relates to the technical field of cutlery and bar utensils, and in particular, an apparatus and method for opening a planar portion of a sealed container.

BACKGROUND

[0003] Traditional opening devices remove a planar upper surface of a sealed container. However, these devices are limited in the types of containers they can open and are constrained in the method of accessing the container contents.

[0004] A recent trend among soda or beverage cans, e.g., beer cans, is to engage the installed tab located at the planar top surface of the can while including a perforated indentation opposite the tab to improve flow of the soda or beverage from the can once opened. However, this conventional technique requires a separate tool, such as a key or other instrument, to puncture the perforated indentation. In addition, for applications in a bar, restaurant, or concession environment, the two-step process of this conventional technique is cumbersome or impractical that inhibits a bartender or concession worker from quickly opening and delivering the container. Moreover, engaging the installed tab and puncturing the perforated indentation may only increase flow of the liquid from the can based on the limited dimensions of the tabbed mouth opening. For example, the perforated indentation constrains the flow of denser liquids. Concurrently, the developmental costs for changing and implementing a perforated indentation design are impractical for the limited amount of target consumers. Moreover, the conventional installed tab, the conventional installed tab and perforated indentation, or traditional opening devices create only a small opening in the container. However, this small opening hinders the user from smelling the full aroma of the container contents. e.g., beer, thereby effecting the user's sense of taste.

[0005] Additionally, traditional opening devices, either manually operated or machine operated, present various access challenges for planar top containers without an engagement tab or mouth. For example, manually operated opening devices use a single blade to remove the entire planar upper surface. These conventional devices require many rotations of a handle or crank to utilize the single blade to cut around the entire circumferential edge of the container. In another example, machine operated opening devices are limited by the speed of the motor during the opening process. Furthermore, both the manually operated and machine operated devices require a user to align and engage the single blade along the circumferential edge before commencing the opening process, which is both time consuming and cumbersome. Moreover, the distance that the shoulder or lip of various containers protrudes outward from the engagement point of the blade can make the opening device ineffective by not allowing the blade to puncture and engage the container surface at an appropriate angle. The shoulder or lip also prevents certain manual devices from fitting on the can and engaging with the rim, while other machine operated devices cause the can to tip at an angle to engage which would cause the contents to spill out during the top removal process.

[0006] Additionally, although traditional manually operated and machine operated devices remove the planar upper surface of a container, these devices leave sharp edges on the container upon removal that are harmful to the user. The contents of such containers subsequently need to be removed to an intermediate container for user safety following the opening process.

SUMMARY

[0007] To overcome the problems described above, embodiments of the present disclosure include a container opener including a handle that when pulled downward creates a sequence of events, for

example, clamping a container, lowering a carriage that causes (i) a barb to puncture a lid of the container, (ii) a blade to puncture the container, and (iii) rotation of the blade and the container with respect to each other to cut off a lid of the container. When the handle is pushed upward, the sequence is reversed such that the carriage is raised, the container is unclamped, and the lid is separated from the barb.

[0008] According to an embodiment of the present disclosure, a container opener includes a clamp that holds a container; a puncturing device that punctures a lid of the container held in the clamp; a blade that cuts the lid from the container while the blade and the container rotate with respect to each other; and a separating device that separates the puncturing device from the lid.

[0009] The container opener can further include a handle, wherein motion of the handle in a first direction rotates the blade and the container with respect to each other. In an aspect, motion of the handle in the first direction forces the puncturing device to puncture the lid. In an aspect, motion of the handle in a second direction causes the separating device to separate the puncturing device from the lid.

[0010] In an aspect, the clamp is configured to hold containers having different diameters.

[0011] The container opener can further include a motor that rotates the blade and the container with respect to each other.

[0012] In an aspect, the blade is a plurality of blades. In an aspect, the plurality of blades and the container rotate at least 120 degrees with respect to each other.

[0013] In an aspect, the blade and the container rotate at least 360 degrees with respect to each other.

[0014] In an aspect, the puncturing device punctures the lid in a central portion of the lid. In an aspect, the puncturing device moves a set distance to puncture the lid no matter a diameter of the container.

[0015] According to another embodiment of the present disclosure, a method of opening a container includes clamping the container; puncturing a lid of the container by a puncturing device; capturing the lid by the puncturing device; rotating a blade and the container with respect to each other to cut the lid from the container; unclamping the container; and separating the puncturing device from the lid.

[0016] In an aspect, motion of a handle in a first direction causes the clamping, the puncturing, and the rotating. In an aspect, motion of the handle in a second direction causes the unclamping, and the separating.

[0017] In an aspect, wherein the rotating is performed by a motor.

[0018] In an aspect, the blade rotates and the container is stationary.

[0019] According to another embodiment of the present disclosure a container opener includes a handle; a tab grabber to grab a tab on a lid of a container; a cutting blade; and an actuating mechanism operatively coupled to the handle, the tab grabber, and the cutting blade; wherein the actuating mechanism is configured to translate a rotational movement of the handle in a first direction to a movement of the tab grabber and the cutting blade in a vertical direction of the container opener such that the tab grabber grabs the tab of the lid of the container held in the container opener and the cutting blade cuts off the lid.

[0020] In an aspect, the actuating mechanism is configured to translate a rotational movement of the handle in a second direction to separate the tab grabber from the lid.

[0021] The above and other features, elements, characteristics, steps, and advantages of the present invention will become more apparent from the following detailed description of preferred embodiments of the present invention with reference to the attached drawings.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

[0022] The following drawings are illustrative of particular embodiments of the present disclosure and therefore do not limit the scope of the present disclosure. The drawings are not to scale and are intended for use in conjunction with the explanations in the following detailed description.

[0023] FIG. 1 is an exploded perspective view of a container opener according to a first embodiment of the present disclosure.

[0024] FIG. 2 is an exploded front view of the container opener of FIG. 1.

[0025] FIG. 3 is a perspective view of a container opener in a home state.

[0026] FIG. 4 is a perspective view of a container opener in an operational state.

[0027] FIG. 5 is a front view of a container opener with a handle up.

[0028] FIG. 6 is a front view of container opener with a handle down.

[0029] FIG. 7 is a left side view of a container opener with a handle up.

[0030] FIG. 8 is a left side view of a container opener with a handle down.

[0031] FIG. 9A is a front view of a container opener with a handle up.

[0032] FIG. 9B is a right side cross-sectional view along C-C of FIG. 9A.

[0033] FIG. 10A is a front view of a container opener with the handle down.

[0034] FIG. 10B is a right side cross-sectional view along D-D of FIG. 10A.

[0035] FIG. 11A is a top view of container opener with the handle up.

[0036] FIG. 11B is a front cross-sectional view along E-E of FIG. 11A.

[0037] FIG. 12A is a top view of a container opener with the handle down and a can in place.

[0038] FIG. 12B is a front cross-sectional view along F-F of FIG. 12A.

[0039] FIG. 13 is a perspective view of a main housing.

[0040] FIG. 14A is a top view of a main carriage and a blade assembly.

[0041] FIG. 14B is a side cross-sectional view along J-J of FIG. 14A.

[0042] FIG. 15A is top view of a main carriage 104 and a blade assembly.

[0043] FIG. 15B is a side cross-sectional view along K-K of FIG. 15A.

[0044] FIG. 16A is a top view of a blade assembly including a can below.

[0045] FIG. 16B is a side cross-sectional view along U-U of FIG. 16A.

[0046] FIG. 17A is a top view of a blade assembly including a can below.

[0047] FIG. 17B is a side cross-sectional view along Y-Y of FIG. 17A.

[0048] FIG. 18A is a top view of a blade assembly including a can below.

[0049] FIG. 18B is a side cross-sectional view along AC-AC of FIG. 18A.

[0050] FIG. 19A is a side view of clamp assembly with jaws open.

[0051] FIG. 19B is a same view as FIG. 19A without the section notation and showing the locations of a can clamp bearing block and a clamp cam slot.

[0052] FIG. 19C is a top cross-sectional view along AD-AD of FIG. 19A.

[0053] FIG. 20A is a side view of a clamp assembly with jaws closed.

[0054] FIG. 20B is a same view as FIG. 20A without the section notation.

[0055] FIG. 20C is a top cross-sectional view along 70E-AE of FIG. 20A.

[0056] FIG. 21A and FIG. 21B are top views of a blade assembly including a can below.

[0057] FIG. 21C is a side cross-sectional view along AF-AF of FIG. 21A.

[0058] FIG. 21D a top cross-sectional view along AG-AG of FIG. 21B.

[0059] FIG. 22A and FIG. 22B are top views of a blade assembly including a can below.

[0060] FIG. 22C is a side cross-sectional view along AH-AH of FIG. 22A.

[0061] FIG. 22D a top cross-sectional view along AJ-AJ of FIG. 22B.

[0062] FIG. 23A and FIG. 23B are top views of the blade assembly including a can below.

[0063] FIG. 23C is a side cross-sectional view along AK-AK of FIG. 23A.

[0064] FIG. 23D a top cross-sectional view along AL-AL of FIG. 23B.

[0065] FIG. 24 is a perspective view of a container opener according to a second embodiment of

the present disclosure.

[0066] FIG. **25** a perspective view of a container opener of FIG. **24** in an operational state.

[0067] FIG. **26** is a front view of a container opener with a handle in the up position.

[0068] FIG. **27** is a front view of a container opener with a handle in the down position.

[0069] FIG. **28** is a right side view of a container opener with a handle in the up position.

[0070] FIG. **29** is a right side view of a container opener with a handle **201** in the down position.

[0071] FIG. **30A** is a top view of container opener with a handle in the up position.

[0072] FIG. **30B** is front cross-section view along section AB-AB of FIG. **30A**.

[0073] FIG. **31A** is a top view of a container opener with a handle in the down position.

[0074] FIG. **31B** is a front cross-section view along section F-F of FIG. **31A**.

[0075] FIG. **32A** is a front view of a container opener with a handle in the up position.

[0076] FIG. **32B** is a side cross-section view along section Y-Y of FIG. **32A**.

[0077] FIG. **33A** is a front view of a container opener with a handle in the down position.

[0078] FIG. **33B** is a side cross-section view along section J-J of FIG. **33A**.

[0079] FIG. **34** is a perspective view of a gearbox assembly and a blade assembly.

[0080] FIG. **35** is an exploded view of a gearbox assembly.

[0081] FIG. **36** shows a larger bevel gear attached to a handle.

[0082] FIG. **37A** is a view of a gearbox assembly with the handle in the up position.

[0083] FIG. **37B** is a side cross-section view of a gearbox assembly taken along section T-T of FIG. **37A**.

[0084] FIG. **38A** is a front view of a gearbox assembly with a handle in the down position.

[0085] FIG. **38B** is a side cross-section view of the gearbox assembly taken along section U-U of FIG. **38A**.

[0086] FIG. **39** is a perspective view of a blade assembly with a can.

[0087] FIG. **40** is an exploded view of a blade assembly

[0088] FIG. **41A** and FIG. **41B** are underside views showing blades in different positions.

[0089] FIG. **42A** is a side view of a blade assembly with a can below.

[0090] FIG. **42B** is a cross-section view taken along section F-F of FIG. **42A**.

[0091] FIG. **42C** is a side view of a blade assembly with a can below.

[0092] FIG. **42D** is a cross-section view taken along section G-G of FIG. **42C**.

[0093] FIG. **42E** is a side view of a blade assembly with a can below.

[0094] FIG. **42F** is a cross-section view taken along section L-L of FIG. **42E**.

[0095] FIG. **43** is a magnified version of the section shown in FIG. **42B**.

[0096] FIG. **44** is a top perspective view of a clamp assembly.

[0097] FIG. **45A** and FIG. **45B** are top views of a clamp assembly with a can.

[0098] FIG. **46** is a top view of a clamp assembly with a can.

[0099] FIG. **47A** and FIG. **47B** are top views of clamps.

[0100] FIG. **48A** is a side view showing clamps open for loading a can.

[0101] FIG. **48B** is a side view showing clamps closed securing a can.

[0102] FIG. **49A** is a front view of a container opener with the handle in the down position.

[0103] FIG. **49B** is a cross-section view taken along section M-M of FIG. **49A**.

[0104] FIG. **50A** is a front view of a container opener with the handle in the down position.

[0105] FIG. **50B** is a cross-section view taken along section R-R of FIG. **50A**.

[0106] FIG. **50C** is a front view of a container opener with the handle in the up position.

[0107] FIG. **50D** is a cross-section view taken along section T-T of FIG. **50C**.

[0108] FIG. **51A** is a front view of a container opener with the handle in the up position.

[0109] FIG. **51B** is a cross-section view taken along section U-U of FIG. **51A**.

[0110] FIG. **52** is an exploded view of the container opener according to the second embodiment of the disclosure.

[0111] FIG. **53** is a top perspective view of a container opener according to a third embodiment of

the present disclosure.

[0112] FIG. **54** is a perspective exploded view of the container opener of FIG. **53**.

[0113] FIG. **55** is a front exploded view of a container opener.

[0114] FIG. **56** is a perspective view of a container opener without a stand.

[0115] FIG. **57** is a perspective view of a container opener including a wall stand.

[0116] FIG. **58** is a perspective view of a container opener in a floor stand.

[0117] FIG. **59** is a perspective view of a container opener with a handle **301** in the down position.

[0118] FIG. **60** is a front view of container opener with a handle in the up position.

[0119] FIG. **61** is a front view of a container opener with a handle in the down position.

[0120] FIG. **62** is a left side view of a container opener with a handle in the up position.

[0121] FIG. **63** is a left side view of a container opener with a handle in the down position.

[0122] FIG. **64A** is a top view of a container opener with a handle in the up position.

[0123] FIG. **64B** is a front cross-section view taken along C-C of FIG. **64A**.

[0124] FIG. **65A** is a top view of a container opener with a handle in the down position.

[0125] FIG. **65B** is a front cross-section view taken along D-D of FIG. **65A**.

[0126] FIG. **66A** is a top view of a container opener with a handle **301** in the up position.

[0127] FIG. **66B** is a right side cross-section view taken along G-G of FIG. **66A**.

[0128] FIG. **67A** is a top view of a container opener with a handle in the down position.

[0129] FIG. **67B** is a right side cross-section view taken along H-H of FIG. **67A**.

[0130] FIG. **68** is a perspective view of a handle and a gear assembly.

[0131] FIG. **69** is a view of a gear train.

[0132] FIG. **70A** is a top view of a handle and a gear assembly with the handle in the up position.

[0133] FIG. **70B** is a right side cross-section view taken along J-J of FIG. **70A**.

[0134] FIG. **71A** is a top view of a handle and a gear assembly with the handle in the down position.

[0135] FIG. **71B** is a right side cross-section view taken along K-K of FIG. **71A**.

[0136] FIG. **72** shows a handle in the down position.

[0137] FIG. **73** shows a handle in the up position.

[0138] FIG. **74** shows a handle in the past-vertical position.

[0139] FIG. **75A** is a top view of a handle and a gear assembly with the handle in the down position.

[0140] FIG. **75B** is a right side cross-section view taken along L-L of FIG. **75A**.

[0141] FIG. **76A** is a top view of a handle and a gear assembly with the handle in the up position.

[0142] FIG. **76B** is a right side cross-section view taken along M-M of FIG. **76A**.

[0143] FIG. **77** is a top perspective view of a blade assembly.

[0144] FIG. **78A** is a top view of a blade assembly.

[0145] FIG. **78B** is a front cross-section view taken along N-N of FIG. **78A**.

[0146] FIG. **79A** is a top view of a blade assembly.

[0147] FIG. **79B** is a front cross-section view taken along R-R of FIG. **79A**.

[0148] FIG. **80A** is a top view of a blade assembly.

[0149] FIG. **80B** is a front cross-section view taken along V-V of FIG. **80A**.

[0150] FIG. **81A** is a top view of a blade assembly.

[0151] FIG. **81B** is a front cross-section view taken along AA-AA of FIG. **81A**.

[0152] FIGS. **82A** is a view of a plunger.

[0153] FIG. **82B** is a top view of a clamp assembly.

[0154] FIG. **83A** is a view of a plunger.

[0155] FIG. **83B** is a top view of a clamp assembly.

[0156] FIG. **84** and FIG. **85** are side views of a clamp assembly.

[0157] FIG. **86** shows an alternative configuration including a motor.

[0158] FIG. **87** is a top perspective view of a container opener with a handle in the up position and

a can in position.

[0159] FIG. **88** is a top perspective view of a container opener with a handle in the down position.

[0160] FIG. **89** is a top view showing a can in position before being pushed into clamps.

[0161] FIG. **90** is a top view showing a can in position while being pushed into clamps.

[0162] FIG. **91** is a front view of FIG. **90**.

[0163] FIG. **92** is a top view showing a can continued to be pushed into clamps.

[0164] FIG. **93** is a top view showing a can pushed all the way into clamps.

[0165] FIG. **94A** is a top view with a can in position all the way into clamps.

[0166] FIG. **94B** is a cross-section view along J-J of FIG. **94A**.

[0167] FIG. **95A** is a top view with a can in position.

[0168] FIG. **95B** is a cross-section view along J-J of FIG. **95A**.

[0169] FIG. **96A** is a top view with a can **410** clamped in position

[0170] FIG. **96B** is a cross-section view along M-M of FIG. **96A**.

[0171] FIG. **97A** is a top view with a can clamped in position and a handle up.

[0172] FIG. **97B** is a cross-section view along C-C of FIG. **97A**.

[0173] FIG. **98A** is a top view with a can clamped in position and a handle down.

[0174] FIG. **98B** is a cross-section view along D-D of FIG. **98A**.

[0175] FIG. **99A** is a side view with a can clamped in position and a handle down.

[0176] FIG. **99B** is a cross-section view along G-G of FIG. **99A**.

[0177] FIG. **100A** is a top view with the can clamped in position and a handle up.

[0178] FIG. **100B** is a cross-section view along E-E of FIG. **100A**.

[0179] FIG. **101A** is a top view with the can clamped in position and with handle up.

[0180] FIG. **101B** is a cross section view along F-F of FIG. **101A**.

[0181] FIG. **102** is a top view of an open clamp.

[0182] FIG. **103** is a top view of a closed clamp.

DETAILED DESCRIPTION

[0183] The following discussion omits or only briefly describes conventional features of container opening devices that are apparent to those skilled in the art. It is noted that various embodiments are described in detail with reference to the drawings, in which like reference numerals represent like parts and assemblies throughout the several views. Reference to various embodiments does not limit the scope of the claims attached hereto. Additionally, any examples set forth in this specification are intended to be non-limiting and merely set forth some of the many possible embodiments for the appended claims. Further, particular features described herein can be used in combination with other described features in each of the various possible combinations and permutations.

[0184] Unless otherwise specifically defined herein, all terms are to be given their broadest reasonable interpretation including meanings implied from the specification as well as meanings understood by those skilled in the art and/or as defined in dictionaries, treatises, etc. It must also be noted that, as used in the specification and the appended claims, the singular forms “a,” “an” and “the” include plural referents unless otherwise specified, and that the terms “includes” and/or “including,” when used in this specification, specify the presence of stated features, elements, and/or components, but do not preclude the presence or addition of one or more other features, steps, operations, elements, components, and/or groups thereof. Moreover, the drawing figures are not necessarily to scale and certain features may be shown exaggerated in scale or in somewhat schematic form in the interest of clarity and conciseness. In the description, relative terms such as “horizontal,” “vertical,” “up,” “down,” “top” and “bottom” as well as derivatives thereof (e.g., “horizontally,” “downwardly,” “upwardly,” etc.) should be construed to refer to the orientation as then described or as shown in the drawing figure under discussion. These relative terms are for convenience of description and are not intended to require a particular orientation. Terms including “inwardly” versus “outwardly,” “longitudinal” versus “lateral” and the like are to be interpreted

relative to one another or relative to an axis of elongation, or an axis or center of rotation, as appropriate. Terms concerning attachments, coupling and the like, such as “connected” and “interconnected,” refer to a relationship wherein structures are secured or attached to one another either directly or indirectly through intervening structures, as well as both movable or rigid attachments or relationships, unless expressly described otherwise. The term “operatively connected” is such an attachment. coupling or connection that allows the pertinent structures to operate as intended by virtue of that relationship.

[0185] The embodiments described herein are directed to opening sealed containers, such as sealed cans, which permit the user to separate a portion or an entire upper planar surface of the container from a pressed outer edge or rim of the container.

[0186] Additionally, by way of example, the embodiments described herein permit the user to safely open and/or remove the upper planar surface of a sealed container with both ease and efficiency. Further, by way of example, the embodiments described herein permit the user to open and/or remove the entire upper planar surface of such container with minimal resulting sharp edges. Moreover, the embodiments described herein permit the user to smell the full aroma of the container contents, e.g., beer. thereby allowing the user to fully enjoy the taste of the container contents.

[0187] FIGS. **1-23** are used to describe a container opener device **100** of a first embodiment of the present disclosure. FIG. **1** is an exploded perspective view and FIG. **2** is an exploded front view showing primary components of the container opener **100**. From top to bottom. FIGS. **1** and **2** show a handle **101**, a rack guide cartridge **102**, a plunger cartridge **103**, a main cartridge **104**, a blade assembly **105**, a barb **106**, a clamp assembly **107**, and a base **108**.

[0188] FIG. **3** is a perspective view of the container opener **100** in a home state. In this position the main carriage **104**, the plunger carriage **103**, the rack guide carriage **102**, and the handle **101** are in their upward positions.

[0189] FIG. **4** a perspective view of the container opener **100** in an operational state. FIG. **4** shows that as the handle **101** is pulled downward, the three carriages (the main carriage **104**, the plunger carriage **103**, the rack guide carriage **102**) lower to create a sequence of events. The sequence is as follows: a can **110** is clamped, the barb **106** punctures the can **110**, a blade punctures the can **110**, and the blade assembly **105** spins at least 360 degrees to cut a lid off the can **110**. This sequence is then reversed when the handle **101** is lifted. These sequences are described in greater detail below.

[0190] FIG. **5** is a front view of the container opener **100** with the handle **101** up. FIG. **5** shows the rack guide carriage **102**, the plunger carriage **103**, the main carriage **104**, and the blade assembly **105**.

[0191] The rack guide carriage **102** lowers through the entirety of the handle stroke. As the rack guide carriage **102** lowers it spins a pinion (located on the main carriage **104**) to spin a gear train and allow the blade assembly **105** to rotate. More details on this can be found with respect to FIG. **14**.

[0192] The plunger carriage **103** is what pushes the main carriage **104** down to its lower position when the handle **101** is pulled down. The plunger carriage **103** mounts a plunger shaft that is used to extend the blade and the barb to puncture the can **110**. There are two springs (see FIG. **15**) between the plunger carriage **103** and the main carriage **104** that sequence these two carriages so that the main housing lowers (to actuate the can clamp) first. The main housing is a subassembly that holds the gear train and the blade assembly **105**. It houses the bearings/shaft that allow the blade assembly **105** to rotate. The main housing raises and lowers with movement of the handle **101**. The movement of the main housing is coupled to the can clamp. When the main housing lowers it closes the clamp and vice versa. Also, the main housing lifts up at the end of the handle stroke to pull the lid out of the can **110** (while the can **110** is still clamped) then unclamps the can **110**.

[0193] Once the main housing bottoms out, the plunger continues to travel, closing the gap

between the plunger carriage **103** and the main carriage **104** so the barb and blade can be extended

[0194] The main carriage **104** is used to mount bearings that allow the blade assembly **105** to spin, a pinion gear and a gear train to spin the blade assembly, and cam followers that extend and retract the can clamp. An upward vertical motion of the main carriage **104** pulls the top of the can **110** out of the body of the can **110** after it is cut and unclamps the can **110**.

[0195] FIG. **6** is a front view of the container opener **100** with the handle **101** down showing that pulling the handle **101** down forces the main carriage **104**, the plunger carriage **103**, and the rack guide carriage **102** together.

[0196] FIG. **7** is a left side view of the container opener **100** with the handle **101** up. FIG. **8** is a left side view of the container opener **100** with the handle **101** down.

[0197] FIG. **9A** is a front view of the container opener **100** with the handle **101** up. FIG. **9B** is a right side cross-sectional view along C-C of FIG. **9A** showing a cam slot **111** in the handle **101**. The cam slot **111** in the handle **101** has two sections. A vertical motion section pushes the plunger carriage **103** down to a set distance. A rotary motion section of the cam slot **111** is concentric with a pivot of the handle **101** which locks the plunger carriage **103** in its lower position and allows the handle **101** to continue to rotate, without moving the plunger carriage **103**, to continue driving the rack guide **102** down to allow the can **110** to spin.

[0198] FIG. **10A** is a front view of the container opener **100** with the handle **101** down. FIG. **10B** is a right side cross-sectional view along D-D of FIG. **10A**. FIG. **10B** show links **112**, the rack guide carriage **102**, and a rack and pinion **113**. The rack guide carriage **102** is driven down by a linkage from the handle cam **111**.

[0199] FIG. **11A** is a top view of the container opener **100** with the handle **101** up. FIG. **11B** is a front cross-sectional view along E-E of FIG. **11A**. FIG. **11B** shows springs **1041** between the plunger cartridge **103** and the main carriage **104**. These springs **1041** are also shown in FIG. **2**.

[0200] FIG. **12A** is a top view of the container opener **100** with the handle **101** down and a can **110** in place. FIG. **12B** is a front cross-sectional view along F-F of FIG. **12A**.

[0201] FIG. **13** is a perspective view of a main housing. The main housing is the larger component that everything is attached to and includes three main components, the main carriage **104** including a housing and guides, a gear train **1042**, and the blade assembly **105**. Guides are bushings inside the main carriage **104** that allow the main housing to slide up and down on vertical guide shafts. The gear train **1042** allows the blade assembly **105** to spin. When a rack gear is pushed down by the handle **101**, the smaller spur gear and horizontal bevel gear are spun. The bevel gear spins the vertical bevel gear which spins the whole blade assembly **105**. The blade assembly **105** is mounted to the main housing through a shaft that goes through the larger bearings in the middle. The blade assembly **105** is attached to a central shaft that goes through the bearings and the vertical bevel gear. A ventral shaft is hollow with a smaller shaft inside. This shaft is connected to the plunger carriage **103** so when the plunger carriage **103** is pushed down, it pushes the smaller internal shaft actuating the blade **115** and the barb **106**.

[0202] FIG. **14A** is top view of the main carriage **104** and blade assembly **105**. FIG. **14B** is a side cross-sectional view along J-J of FIG. **14A** showing the plunger shaft **114**, the barb **106**, and blade **115**. FIG. **14B** is showing a blade arm **119** in its initial unengaged position.

[0203] FIG. **15A** is top view of the main carriage **104** and the blade assembly **105**. FIG. **15B** is a side cross-sectional view along K-K of FIG. **15A**. FIG. **15B** is showing the blade arm **119** after it has been pressed down to engage the blade **115** into the can **110**.

[0204] FIGS. **16-19** are used to further describe the operation of the container opener **100**. FIG. **16A** is a top view of the blade assembly **105** including a can **110** below the blade assembly **105**. FIG. **16B** is a side cross-sectional view along U-U of FIG. **16A** and includes Detail M that is a closer scaled portion. FIG. **16B** shows that the plunger shaft **114**, which is tied into the plunger carriage **103** pushes down on a blade spring **116** which pushes down a spring block **118** which pushes down a blade arm **119**. The blade arm **119** is a lever to rotate the blade **115**

counterclockwise to extend the blade **115** into the can **110**. The barb **106** is connected directly from the blade spring shaft **117** to the plunger shaft **114**. The blade spring **116** does not affect the barb **106**. When the plunger shaft **114** is lowered, the blade spring **116** pushes the spring block **118** down in turn rotating the blade arm **119** to extend the blade **115** to contact the inner rim of the can **110**. At the same time the barb **106** is forced down in the small circle in the center of the tab of the can **110**. The widest diameter of the barb **106** punctures through the wall of the lid of the can **110**.

[0205] FIG. **17A** is a top view of the blade assembly **105** including a can **110** below. FIG. **17B** is a side cross-sectional view along Y-Y of FIG. **17A** and includes Detail W that is a closer scaled portion. FIG. **17B** shows that the plunger shaft **114** is lowered to its final position. This sets the barb **106** to its fully extended position. Once the blade **115** contacts the inner rim of the can **110**, as shown in FIG. **16B**, the blade arm **119** is now fixed. Forcing the blade **115** outward helps to minimize any exposed sharp edges on the can **110** after the lid is cut off. To permit for the plunger shaft **114** to continue travelling to its final position, the spring block **118** needs to slide relative to the blade spring shaft **117** to define a gap **120** between the spring block **118** and a step on the blade spring shaft **117**. The spring block **118** compresses the blade spring **116**, preloading the blade arm **119** to ensure a proper amount of blade **115** contact force. The blade **115** hits the edge of the can **100**, so the spring block **118** cannot travel any further down. The blade spring shaft **117** then keeps sliding through the spring block **118** to account for any tolerance on the can **110** and its position. Defining the gap **120** and the compression of the blade spring **116** plays a role in the container opener's ability to cut different diameter cans **110**. The larger the diameter of the can is, the smaller the gap **160** and vice versa for smaller diameter cans. This is important because it allows the plunger shaft **114**/barb **106** to have a fixed stroke regardless of the actual diameter of the can **110**. Detail W shows a larger gap **160** defined due to a smaller diameter of can **110**.

[0206] FIG. **18A** is a top view of the blade assembly **105** including a can **110** below. FIG. **18B** is a side cross-sectional view along AC-AC of FIG. **18A** and includes Detail AB that is a closer scaled portion. FIG. **18B** is similar to that of FIG. **17B** but is used to show that the container opener **100** can accommodate different diameter cans **110**. Detail AB shows that a smaller gap **120** is defined due to a larger can diameter.

[0207] FIGS. **19** and **20** are used to show the operation of the clamp assembly **107** as it clamps a can **110**. FIG. **19A** is a side view of the clamp assembly **107** with the jaws **121** open, a gap between the two jaws **121** is present. FIG. **19B** is a same view as FIG. **19A** without the section notation and showing the locations of a can clamp bearing block **122** and a clamp cam slot **123**. FIG. **19C** is a top cross-sectional view along AD-AD of FIG. **19A**. FIG. **20A** is a side view of the clamp assembly **107** with the jaws **121** closed, there is no gap between the jaws **121**. FIG. **20B** is a same view as FIG. **20A** without the section notation. FIG. **20C** is a top cross-sectional view along AE-AE of FIG. **20A**. FIGS. **19** and **20** show that as the main carriage **104** is driven down by the handle **101**, the vertical motion of the can clamp bearing block **122** drives the clamp cam slot **123** inwards, thus moving the jaws **121** inward and securely clamping the can **110**.

[0208] FIGS. **21-23** show an alternate blade assembly **1050** according to another aspect of the disclosure. FIGS. **21A**, **21B**, **22A**, **22B**, **23A**, and **23B** are top views of the blade assembly **1050** including the can **110** below. FIG. **21C** is a side cross-sectional view along AF-AF of FIG. **21A**. FIG. **21D** a top cross-sectional view along AG-AG of FIG. **21B**. FIG. **22C** is a side cross-sectional view along AH-AH of FIG. **22A**. FIG. **22D** a top cross-sectional view along AJ-AJ of FIG. **22B**. FIG. **23C** is a side cross-sectional view along AK-AK of FIG. **23A**. FIG. **23D** a top cross-sectional view along AL-AL of FIG. **23B**.

[0209] In this aspect, a blade arm **1190** is sprung outward by two blade springs **1160**. When the plunger shaft **114** is lowered, it pushes a cam block **124** down, releasing the blade arm **1190** so the blade **115** can contact the can **110**. With respect to FIG. **23**, once the blade **115** makes contact with the can **110**, the blade arm **1190** is now disconnected from the cam block **124** allowing the blade springs **1160** to apply the necessary contact force to the blade **115**. This also allows the plunger

carriage **103** to continue moving downward so the barb **106** can reach its final position. When the plunger shaft **114** is raised, the cam block **124** pushes on the cam slot on the blade arm **1190** to retract the blade **115**, compressing the blade springs **1160**, and retracting the barb **106**. FIG. **23** shows that because the blade arm **1190** is sprung outward and is independent to the final position of the plunger shaft **114**, this configuration also can accommodate different diameters of cans **110**. [0210] FIGS. **24-52** are used to describe a container opener **200** of a second embodiment of the present disclosure. FIG. **24** shows the container opening **200** includes a handle **201**, clamp shafts **202**, a plunger carriage **203**, a blade assembly **205**, a clamp assembly **207**, and a base **208**. In an operation sequence, as the handle **201** is pulled down, the plunger carriage **203** lowers, driving the clamp shafts **202** downward. The clamp shafts **202** actuate to clamp and secure a can **210**, a barb **203** punctures the can **210**, a blade **215** punctures the can **210**, and the blade assembly **205** pins at least **120** degrees to cut the top off the can **210**. This sequence is then reversed when the handle **201** is lifted.

[0211] FIG. **25** shows the final state as the handle **201** is lowered to its down position. At this stage the can **210** is secured in the clamp assembly **207**, the blade **215** and barb **203** have punctured the can **210**, and the blade assembly **205** has rotated at least 120 degrees.

[0212] FIG. **26** is a front view of the container opener **200** with the handle **201** in the up position. FIG. **27** is a front view of the container opener **200** with the handle **201** in the down position. Here, the clamp assembly **207** is closed and tightly holds the can **210**.

[0213] FIG. **28** is a right side view of the container opener **200** with the handle **201** in the up position. FIG. **29** is a right side view of the container opener **200** with the handle **201** in the down position.

[0214] FIG. **30A** is a top view of the container opener **200** with the handle **201** in the up position. FIG. **30B** is front cross-section view along section AB-AB of FIG. **30A**. FIG. **31A** is a top view of the container opener **200** with the handle **201** in the down position. FIG. **31B** is front cross-section view along section F-F of FIG. **31A**.

[0215] FIG. **32A** is a front view of the container opener **200** with the handle **201** in the up position. FIG. **32B** is side cross-section view along section Y-Y of FIG. **32A**. FIG. **33A** is a front view of the container opener with the handle **201** in the down position. FIG. **33B** is side cross-section view along section J-J of FIG. **33A**. FIG. **32** shows a cam slot **211** in the handle **201** can include two sections: a vertical motion portion pushes the plunger carriage **203** down to a set distance; and during rotary motion the position of the cam slot **211** is concentric with the pivot of the handle **201** which locks the plunger carriage **203** in its lower position and allows the handle **201** to continue to rotate, without moving the plunger carriage **203**, to drive a gear train, in turn spinning the blade assembly **205**, as described in more detail below.

[0216] FIG. **34** is a perspective view of a gearbox assembly **204** and the blade assembly **205**. FIG. **35** is an exploded view of the gearbox assembly **204**. FIG. **36** shows a larger bevel gear **2041** attached to the handle **201**, so when the handle **201** is pulled downward, the larger bevel gear **2041** rotates a smaller bevel gear **2042**, which then rotates a larger spur gear **2043**, which then rotates a smaller spur gear **2044**. This gear train causes the blade assembly **205** to rotate with the motion of the handle **201**. The rotating blades will cut the lid off the can **210**.

[0217] FIG. **37A** is a view of the gearbox assembly **204** with the handle **201** in the up position. FIG. **37B** is a side cross-section view of the gearbox assembly **204** taken along section T-T of FIG. **37A**. As the handle **201** is pulled down, a roller **2045** slides in the cam slot **211**. The first part of the motion causes a downward motion of a plunger block **2031** of the plunger carriage **203**, then the cam slot **211** is concentric with the pivot, so the blade assembly **205** keeps rotating without any more downward motion. FIG. **38A** is a front view of the gearbox assembly **204** with the handle **201** in the down position. FIG. **38B** is a side cross-section view of the gearbox assembly **204** taken along section U-U of FIG. **38A**. As shown, when the handle **201** is at horizontal, the roller **2045** has travelled its full distance in the cam slot **211**.

[0218] FIG. 39 is a perspective view of the blade assembly 205 with a can 210. FIG. 40 is an exploded view of the blade assembly 205 showing that three blades 215 rotate out to make contact with the can 210 and cut the lid off.

[0219] FIGS. 41A and 41B are underside views showing the blades 215 in different positions. When the handle 201 is in the up position, the blades 215 are retracted and inboard as shown in FIG. 41A. As the handle 201 is pulled, the blades 215 travel radially outwards (see FIG. 41B) to contact the can 210. The retracted blades 215 provide enough space that the can 210 can be inserted into the clamp assembly 207 before the blades 215 move outwardly and cut into the sidewall of the can 210.

[0220] FIGS. 42A-42F are used to show a cycle operation of one representative blade 215. All blades 215 operate in the same manner. FIGS. 42A, 42C, and 42E are side views of the blade assembly 205 with a can 210 below. FIG. 42B, taken along section F-F of FIG. 42A shows that a blade roller 2051 starts in a detent of a cam surface of a blade cam 2052, as the handle 201 is pulled down the blade roller 2051 rolls down the cam surface of the blade arm 2052 until the blade 215 makes contact with the can 210. FIG. 42D, taken along section G-G of FIG. 42C shows that the blade 215 is forced outwards by a blade spring 2054, so when the blade 215 touches the can 210, the blade roller 2051 leaves the surface of the blade arm 2052 and can continue traveling downwards. The blade spring 2054 provides enough force for the blade 215 to penetrate the can 210 and cut off the lid of the can 210 as the blade assembly 205 rotates. Forcing the blade 215 outward helps to minimize any exposed sharp edges on the can 210 after the lid is cut off. FIG. 42F, taken along section L-L of FIG. 42E shows that by allowing the blade roller 2051 to leave the surface of the blade cam 2052, the blades 215 can cut different diameter cans 210. The blades 215 will rotate further outwards before reaching the inner diameter (ID) of a larger can 210). Note, that there are three sets of blades 215 that simultaneously follow the sequence described above. FIG. 42D includes a can 210 with a smaller diameter can rim than that shown in FIG. 42F. These figures show that container opener 200 can cut different diameters of cans. The blade 215 travels out to different positions for the different can sizes, but the blade roller 2051 travels to the same location.

[0221] FIG. 43 is a magnified version of the section shown in FIG. 42B. FIG. 43 shows that the downward motion of the plunger shaft 214 of the plunger carriage 203 also extends the barb 206 towards the can 210 to puncture in the small circle in the middle of the tab on the lid of the can 210. Because the blade roller 2051 can separate from the surface of the blade cam 2052, the plunger shaft 214 and barb 206 can extend a set distance no matter the diameter of the can 210.

[0222] FIGS. 44-46 are used to describe the clamp assembly 207. FIG. 44 is a top perspective view of the clamp assembly 207. FIG. 44 shows that the clamp assembly 207 can include a clamp 2071, a link 2072, a link attachment block 2073, a urethane block 2074, a clamp spring 2075, and a lid flap 2076. Two clamps 2071 are shown. In operation, the clamps 2071 start forced outwards by the clamp spring 2075 (see FIG. 45A), then, after a can 210 is inserted and when the handle 201 is pulled downward, the vertical motion of the plunger shaft 214 causes the link attachment blocks 2073 to move downward. This drives the links 2072 down, which causes the clamps 2071 to move inward to clamp the can 210 (see FIG. 45B). The clamps 2071 have a V-shape to be able to clamp and center a variety of can diameters. This is important so that the center of the can 210 is always in the same location within the container opener 200 regardless of its diameter. The urethane blocks 2074 on the inside of the clamps 2071 provide additional friction to prevent the can 210 from rotating when the blade assembly 205 is cutting the can 210. FIG. 46 shows the clamp assembly 207 clamping a larger diameter can 210.

[0223] FIGS. 47A and 47B show the lid flaps 2076 as a portion of the clamp assembly 207. The lid flaps 2076 are located on a lower portion of the clamp assembly 207 and are angled to direct a cutoff lid to slide in a different direction. During the reject (upward) stroke of the handle 201, the can 210 has already been removed (without its lid), but the lid is still stuck on the barb 206. When the handle 201 travels back up for the reject, the lid flaps 2076 close, so when the barb 206 is

pulled through the lid, the lid hits the closed lid flaps **2076** and slides into a collection bin at the rear of the container opener **200**. Before the reject, the lid flaps **2076** are open (see FIG. **47A**) so that the can **210** is able to be removed. During the reject, the lid flaps **2076** close (see FIG. **47B**) so the lid will slide back to the collection bin. This operation is shown in greater detail with respect to FIGS. **48-51**.

[0224] FIG. **48A** is a side view showing the clamps **2071** open for loading the can **210**. FIG. **48B** is a side view showing the clamps **2071** closed securing the can **210**. Vertical shafts **2077** are driven down by the plunger block to provide motion to close the clamps **2071**.

[0225] FIG. **49A** is a front view of the container opener **200** with the handle **201** in the down position. FIG. **49B** is taken along section M-M of FIG. **49A**. With the handle **201** in the up position, the barb **206** begins above the top of the can **210**, and during the down stroke, the barb **206** punctures the lid of the can **210**. After the can **210** is cut, the lid stays on the barb **206** and gets pulled out of the can **210**. After the can **210** is removed, the lid is rejected from the barb **206** during the reject (upward) stroke of the handle **201**.

[0226] FIG. **50B**, taken along section R-R of FIG. **50A** of the container opener **200** with the handle **201** in the down position, shows the barb **206** inside of the can **210**. FIG. **50D**, taken along section T-T of FIG. **50C** with the handle **201** in the up position, shows that the barb **206** has retracted and the lid **2101** is falling onto closed lid flaps **2076**. FIG. **51B**, taken along section U-U of FIG. **51A** shows that the lid **2101** has hit the lid flaps **2076** and is sliding toward the back of the container opener **200** to be collected.

[0227] FIG. **52** is an exploded view of the container opener **200** according to the second embodiment of the disclosure showing the handle **201**, the clamp shafts **202**, the plunger carriage **203**, the gear assembly **204**, the blade assembly **205**, the clamp assembly **207**, and the base **208** along with a can **210**.

[0228] FIGS. **53-86** are used to describe a container opener **300** of a third embodiment of the present disclosure. FIG. **53** shows the container opener **300** can include a handle **301**, a plunger carriage **303**, clamp shafts **302**, a blade assembly **305**, a clamp assembly **307**, and a stand **308**. A can **310** is also shown. In an operation sequence, as the handle **301** is pulled down, the plunger carriage **303** lowers, driving the clamp shafts **302** downward. The clamp shafts **302** actuate to clamp and secure the can **310**, a barb **306** punctures a lid of the can **310**, a blade **315** punctures the can **310**, and the blade assembly **305** spins at least 360 degrees to cut the top off the can **310**. This sequence is then reversed when the handle **301** is lifted.

[0229] FIG. **54** is a perspective exploded view of the container opener **300** of FIG. **53**. The container opener **300** can be mounted on a table/bar top, on the floor, or on the wall. The stand **308** shown allows for the container opener **300** to sit on a bar top. The container opener **300** has a universal mounting method for any type of base or stand needed.

[0230] FIG. **55** is a front exploded view of the container opener **300** including a gearbox assembly **304**, a blade assembly **305**, sidewall extrusions **309**, and a clamp assembly **307**. A clamped can **310** is also shown.

[0231] FIG. **56** is a perspective view of the container opener **300** without the stand **308**. FIG. **57** is a perspective view of the container opener **300** including a wall stand **313** for mounting the container opener **311** to a wall. As mentioned, the container opener **300** can be mounted in a variety of ways. The same mounting features are used on the container opener **300** so it can be mounted to different stands or using different methods to permit the container opener **300** to be easily switched between mounting techniques. FIG. **58** is a perspective view of the container opener **300** in a floor stand **312**. A bin **3121** can be placed under the container opener **300** to collect the lids as they are removed from the cans **310**.

[0232] FIG. **59** is a perspective view of the container opener **300** with the handle **301** in the down position. FIG. **60** is a front view of the container opener **300** with the handle **301** in the up position. FIG. **61** is a front view of the container opener **300** with the handle **301** in the down position. FIG.

62 is a left side view of the container opener **300** with the handle **301** in the up position. FIG. **63** is a left side view of the container opener **300** with the handle **301** in the down position.

[0233] FIGS. **64-85** are used to describe the operation of the container opener **300**. FIG. **64A** is a top view of the container opener **300** with the handle **301** in the up position. FIG. **64B** is a front cross-section view taken along C-C of FIG. **64A**. As the handle **301** rotates, the plunger carriage **303** moves vertically downwards and pushes clamps **3071** of the clamp assembly **307** inward to tightly clamp the can **310**. The plunger carriage **303** pushes on a plunger shaft **314** which pushes a barb **306** into the lid of the can **310**. This allows for removal of the lid after the lid is cut away from the can **310**. A plunger shaft housing **3141** pushes on a slot in a blade arm **319** which rotates the blades into the wall of the can **310**.

[0234] FIG. **65A** is a top view of the container opener **300** with the handle **301** in the down position. FIG. **65B** is a front cross-section view taken along D-D of FIG. **65A**. The plunger carriage **303** also provides vertical motion for the barb **306** to puncture the lid top before the cutting begins. After the cutting has completed and the lid is separated from the can body, the barb **306** retracts and removes the lid from the can **310**. The open can **310** is then removed. Then, during the reject (upward) stroke when the handle **301** goes upwards past vertical, the lid drops below.

[0235] FIG. **66A** is a top view of the container opener **300** with the handle **301** in the up position. FIG. **66B** is a right side cross-section view taken along G-G of FIG. **66A**. The cam slot **311** of the handle **301** has two sections: a vertical motion portion pushes the plunger carriage **303** down to a set distance, and a rotary motion portion of the cam slot **311** that is concentric with the pivot of the handle **301** which locks the plunger carriage **303** in its lower position and allows the handle **301** to continue to rotate without moving the plunger carriage **303** to drive a gear train of the gear assembly **304** in turn spinning the blade assembly **305**.

[0236] FIG. **67A** is a top view of the container opener **300** with the handle **301** in the down position. FIG. **67B** is a right side cross-section view taken along H-H of FIG. **67A**.

[0237] FIG. **68** is a perspective view of the handle **301** and gear assembly **304**. FIG. **69** is a view of a gear train **3041** (within the dotted circle) and the blade assembly **305**. The gear train **3041** provides rotational motion to the blade assembly **305**. A first set of spur gears **3042** transfers to a set of bevel gears **3043** to rotate the blade assembly **305**. The first set of spur gears **3042** transmit the handle torque to the bevel gears **3043** through a gear ratio. The bevel gears **3043** change the direction of motion. The bevel gears **3043** then transmit torque through a second set of spur gears **3044** (with a gear ratio) to provide rotation of the blade assembly **305**.

[0238] FIG. **70A** is a top view of the handle **301** and gear assembly **304** with the handle **301** in the up position. FIG. **70B** is a right side cross-section view taken along J-J of FIG. **70A**. FIG. **71A** is a top view of the handle **301** and gear assembly **304** with the handle **301** in the down position. FIG. **71B** is a right side cross-section view taken along K-K of FIG. **71A**. The cam slot **311** in the handle **301** creates the plunger carriage's **303** vertical motion, then the cam slot **311** becomes concentric with the pivot, so the vertical motion stops, but the rotational motion continues to spin the gears, spinning the blade assembly **305** to complete the cut of the lid from the can **310**.

[0239] FIGS. **72-76** are used to describe how a cutoff lid is rejected. FIGS. **72-74** are left side views of the handle **301** and gear assembly **304**. FIG. **72** shows the handle **301** in the down position. FIG. **73** shows the handle **301** in the up position. FIG. **74** shows the handle **301** in the past-vertical position. A slot **3042** in the gear assembly **304** can include a detent **3043** to keep the handle **301** in an upright position. When a user retracts the handle **301**, the motion will hit this detent **3043** to impede upward motion of the handle **301** before continuing to the reject phase of the motion, which is the past-vertical position. There is a spring **3012** inside a cam handle **3011** (best viewed in FIG. **70B**) that provides outwards force to keep a roller in the detent **3043**. FIG. **74** shows that the handle **301** must be continued to be pushed backward (past-vertical) to reject the lid of the can **310**.

[0240] FIG. **75A** is a top view of the handle **301** and gear assembly **304** with the handle **301** in the

down position. FIG. 75B is a right side cross-section view taken along L-L of FIG. 75A. FIG. 76A is a top view of the handle 301 and gear assembly 304 with the handle 301 in the up position. FIG. 76B is a right side cross-section view taken along M-M of FIG. 76A. The plunger carriage 303 provides the downward force via the downward handle motion to push the barb 306 into the top of the can 310. The barb 306 is guided on a separate shaft with a light spring to allow for the barb 306 to lag. The lag is necessary to be able to remove the can before the cutoff lid rejects back into the drink.

[0241] FIGS. 77-81 are used to describe the blade assembly 305. FIG. 77 is a top perspective view of the blade assembly 305. FIG. 78A is a top view of the blade assembly 305. FIG. 78B is a front cross-section view taken along N-N of FIG. 78A with a scaled version of Detail P. As the plunger shaft 314 is lowered, it pushes the blade spring block 318 down, pushing the blade springs (not shown) that push the blade bushing block 316. The blade bushing block 316 then pushes down on the blade arm 319 (through a cam in a slot) to rotate the blade arm 319 counterclockwise extending the blade 315 toward the inner rim of the can 310. The blade bushing block 316 also extends the barb 306 towards the can 310.

[0242] FIG. 79A is a top view of the blade assembly 305. FIG. 79B is a front cross-section view taken along R-R of FIG. 79A with a scaled version of Detail U. Once the blade 315 contacts the can 310, the blade arm 319 is now locked into position, but the plunger shaft 314 must continue travelling downward to force the barb 306 to puncture the lid of the can 310. For this to happen, the blade springs can compress allowing the extra travel needed for the plunger shaft 314. This compression also provides a set range of contact force on the blade 315 to accommodate for tolerances within components of the container opener 300 or tolerances of the diameter of the can 310. At the end of the stroke, the barb 306 punctures the center of the can 310 and continues to depress the lid of the can 310 to create downward pressure that helps separate the lid from the can body after it has been cut. Once the blade assembly 305 has spun 360+ degrees the can lid is separated from the can body and the can lid is now resting on a lip of the barb 306.

[0243] FIG. 80A is a top view of the blade assembly 305. FIG. 80B is a front cross-section view taken along V-V of FIG. 80A with a scaled version of Detail Y. With the can lid 3101 severed from the can body, the upward sequence now takes place. As the plunger shaft 314 retracts due to moving the handle 301 upward, the blade springs decompress and the blade spring block 318 bottoms out to pull the blade bushing block 316 up. This rotates the blade arm 319 clockwise to retract the blade 315 and the barb 306. This motion then pulls the can lid 3101 out of the can body until the can lid 3101 hits the blade bottom cover 317.

[0244] For the top level sequence to occur, the can clamp must release the can 310 before the lid 3101 is removed from the barb 306 to avoid the lid 3101 being dropped in the open can 310. This means the handle 301 would need to return to its vertical position which would return the blade assembly 305 to its state shown in FIGS. 78A and 78B. However, because the can lid 3101 is sitting on the barb 306 the thickness of the can lid 3101 now prevents the barb 306 from reaching its “home” position. In order to address this, the barb 306 is attached to a shaft 3061 with a light spring. This spring allows the can lid 3101 to “lag” and create a gap GAP between the barb 306 and the blade bushing block 316. This lag allows the blade bushing block 316/plunger shaft 314 to return to their home positions correlating with the vertical position of the handle 310. If this lag was not present, the handle 310 would return to its vertical position and force the barb 306 through the can lid releasing the can lid 3101 into the open can 310 below the can lid 3101.

[0245] FIG. 81A is a top view of the blade assembly 305. FIG. 81B is a front cross-section view taken along AA-AA of FIG. 81A with a scaled version of Detail AC. At this stage, the can 310 has been removed, the can lid 3101 is pinned between the step on the barb 306 and the blade bottom cover 317. The handle 301 is then rotated to the past-vertical position to the “reject” position. As this happens, the plunger shaft 314/blade bushing block 316 move upwards. This causes the barb shaft 3061 to continue to extend (compressing the spring) until the head of the barb shaft 3061

bottoms out on the step **3161** in the blade bushing block **316**. The lagging of the barb **306** is now completed and the remainder of the reject stroke of the handle **301** pulls the barb **306** through the can lid **3101**, ejecting the can lid **3101** from the barb **306**, shown in FIG. **81B**.

[0246] FIGS. **82-85** are used to describe the clamp assembly **307**. FIGS. **82A** and **83A** are views of a plunger **3031** of the plunger carriage **303** and clamp assembly **307**. FIG. **82B** and **83B** are top views of the clamp assembly **307**. The plunger's **3031** downwards motion forced by handle **301** also pushes a roller in a clamp cam arm **322** that closes the clamps **321**. A profile of a clamp cam slot **323** in the clamp cam arm **322** closes the clamps **321** quickly before the barb **306** or blade **315** punctures the can **310**. During the remainder of the downward stroke, the clamps **321** are kept closed by a vertical portion of the clamp cam slot

[0247] FIGS. **84** and **85** are side views of the clamp assembly **307**. At the "home position", when the handle **301** is vertical, there is enough space between the clamps **321** and the can **310** to comfortably insert the can **310** in the container opener **300**, shown in FIG. **84**. On the downward stroke, the clamps **321** would then close around the can **310** to hold it in place before the barb **306** or the blade **315** punctures the can lid, shown in FIG. **85**. A urethane strip **3074** provides friction to keep the can **310** from rotating during cutting. Over time, it is to be expected that an edge that a lip of the can **310** sits on will wear down. Therefore, there is a can insert **3075** inside of the clamps **321** that can be easily replaced.

[0248] In another aspect, FIG. **86** shows an alternative configuration where instead of gears that rotate the blade assembly **305**, there is a motor **324** that is operated by a switch **325** that can be a push button. The switch **325** can be depressed when the plunger carriage **303** is lowered during the downward stroke of the handle **301**. The motor **324** can be connected to gearing inside a housing. The cutting and removal of the can lid is identical to what is described above.

[0249] FIGS. **87-101** are used to describe a container opener **400** of a fourth embodiment of the present disclosure where the container opener **400** includes an electric motor.

[0250] FIG. **87** is a top perspective view of the container opener **400** with the handle **401** in the up position and a can **410** in position. FIG. **88** is a top perspective view of the container opener **400** with the handle **401** in the down position. FIGS. **89-101** are used to describe operation of the container opener **400**.

[0251] FIG. **89** is a top view showing the can **410** in position before being pushed into clamps **421**. At this stage, the clamps **321** and clamp flaps **4076** are sprung inwards to hardstops. In FIG. **90**, the can **410** has engaged the clamps **421**. The can **410** pushes the clamps **421** outward and compresses the clamp springs **4211**. As shown, there can be two clamp springs **4211** per clamp **421**. The clamp springs **4211** provide equal pressure on the left and right clamps **421** keeping the can **410** centered.

[0252] FIG. **91** is a front view of FIG. **90** showing how the can **410** interacts in the clamps **421**. There is a channel **4212** cut in each of the can clamps **421** that permit the can rim to rest on the bottom of the channel **4212**. The wall at the bottom of the channel **4212** pushes against the can body just under the can rim. FIG. **92** is a top view showing the can **410** continued to be pushed forward causing the can flaps **4076** to push outward compressing their springs **4077**. FIG. **93** shows the can **410** after being pushed all the way into the clamps **421** until the can **410** reaches a drive wheel **413**. At this stage, the can flaps **4076** are sprung outward to hook around a side of the can **410**, keeping the can **410** temporarily in place. The can clamps **421** are sprung inward holding the can **410** centered. The can clamps **421** secure the can **410** during the cut to prevent the can **410** from tipping while being cut.

[0253] FIG. **94A** is a top view with the can **410** in position and FIG. **94B** is cross-section view along J-J of FIG. **94A** showing the can **410** against the drive wheel **413**. The drive wheel **413** can include a channel cut into it that allows the rim of the can **410** to rest on the bottom of the channel. There is also a friction surface that rests against the outer rim of the can **410**. This will provide the necessary friction to spin the can when the blade applies a force to the inside of the rim.

[0254] FIG. **95A** is a top view with the can **410** in position and FIG. **95B** is cross-section view

along J-J of FIG. 95A showing the can **410** against the drive wheel **413**. As shown in Detail N, the drive wheel **413** has a channel **4131** defined into it that allows the rim of the can **410** to rest on the bottom of the channel **4131**. There is also a friction surface **4312** forcing against the outer rim of the can **410**. This will provide the necessary friction to spin the can **410** when the blade applies a force to the inside of the rim.

[0255] FIG. **96A** is a top view with the can **410** clamped in position and FIG. **96B** is a cross-section view along M-M of FIG. **96A** showing the can **410** against the drive wheel **413**. This is showing how the can is resting in the channels **4212** in the can clamps **421**.

[0256] FIG. **97A** is a top view with the can **410** clamped in position and the handle **401** up. FIG. **97B** is cross section view along C-C of FIG. **97A**. FIG. **97B** shows how the blade **415** and a tab grabber **406** are mounted to the handle **401** and are moved away from the can **410** when the handle **401** is up. The handle **401** is held up by a spring **4011** at the rear of the handle **401**. A lid remover **404** is also retracted with a spring **4041**. There is also an electrical switch **425** below the handle **401** that actuates the motor **424** that spins the drive wheel **413**. At this stage, the switch **425** is not engaged.

[0257] FIG. **98A** is a top view with the can **410** clamped in position and the handle **401** down. FIG. **98B** is cross section view along D-D of FIG. **98A**. At this stage, as the handle **401** is pulled down, the blade **415** is engaged with the inner rim of the can **410**. This forces the can **410** against the drive wheel **413**. The force applied from the blade **415** creates enough friction between the can rim and the drive wheel friction surface **4312** to spin the can **410**. The tab grabber **406** is also pushed against the top of the can **410**. When the handle **401** comes down, this action engages the electrical switch **425**. This provides power to the motor **424** to spin the drive wheel **413** and the can **410**.

[0258] FIG. **99A** is a side view with the can **410** clamped in position and the handle **401** down. FIG. **99B** is cross section view along G-G of FIG. **99A**. When the electrical switch is engaged, the motor turns the drive wheel **413**, in turn rotating the can **410**. During the rotation of the can **410**, the tab of the can **410** turns with the can **410** until it is forced on top of one of the feet **4062** of the tab grabber **406**. Once the tab of the can **410** reaches the vertical wall **4061** of the tab grabber **406**, the tab is hooked and does not continue to spin. The can tab then remains stationary while the can **410** continues to spin to finish cutting the can lid.

[0259] FIG. **100A** is a top view with the can **410** clamped in position and the handle **401** up. FIG. **100B** is cross section view along E-E of FIG. **100A**. Once the can lid **4101** is cut, the user would then lift up the handle **401**. This would disengage the electrical switch **425** turning off the motor **424**. Because the can tab is hooked on the tab grabber **406**, the can lid **4101** will be removed with the upward action of the handle **401**. This action also retracts the blade **415** away from the can **410**.

[0260] FIG. **101A** is a top view with the handle **401** up. FIG. **101B** is cross section view along F-F of FIG. **101A**. After the handle **401** is lifted upward, the can **410** is then pulled out of the clamps **421** in the reverse sequence as described above. After that, the user would push on the lever of the lid remover **404** to eject the can lid **4101** off the tab grabber **406** ending the operation sequence.

[0261] FIG. **102** is a top view of another configuration of a can clamp **500** shown open. FIG. **103** is a top view of the clamp **500** closed. A left clamp **502** can be manually closed by rotating a handle **501** clockwise by a user. The left clamp **502** pulls a right clamp **503** closed through a clamp spring **504**. The left clamp **502** is held closed by a latch **505** and the right clamp **503** is held closed with the clamp spring **504**. The clamp spring **504** ensures that a can is securely clamped between the left clamp **502** and the right clamp **503** when the clamp **500** is closed. The clamp **500** can be unlocked by a roller on a handle of a container opener to which the clamp **500** has been incorporated. The roller on the handle of the container opener can hit a flag that rotates a roller **506** to open the latch **505**.

[0262] It should be understood that the foregoing description is only illustrative of the present invention. Various alternatives and modifications can be devised by those skilled in the art without

departing from the present invention. Accordingly, the present invention is intended to embrace all such alternatives, modifications, and variances that fall within the scope of the disclosure.

Claims

1. A container opener comprising: a clamp that holds a container; a puncturing device that punctures a lid of the container held in the clamp; a blade that cuts the lid from the container while the blade and the container rotate with respect to each other; and a separating device that separates the puncturing device from the lid.
2. The container opener of claim 1, further comprising: a handle, wherein motion of the handle in a first direction rotates the blade and the container with respect to each other.
3. The container opener of claim 2, wherein motion of the handle in the first direction forces the puncturing device to puncture the lid.
4. The container opener of claim 2, wherein motion of the handle in a second direction causes the separating device to separate the puncturing device from the lid.
5. The container opener of claim 1, wherein the clamp is configured to hold containers having different diameters.
6. The container opener of claim 1, further comprising a motor that rotates the blade and the container with respect to each other.
7. The container opener of claim 1, wherein the blade is a plurality of blades.
8. The container opener of claim 7, wherein the plurality of blades and the container rotate at least 120 degrees with respect to each other.
9. The container opener of claim 1, wherein the blade and the container rotate at least 360 degrees with respect to each other.
10. The container opener of claim 1, wherein the puncturing device punctures the lid in a central portion of the lid.
11. The container opener of claim 5, wherein the puncturing device moves a set distance to puncture the lid no matter a diameter of the container.
12. A method of opening a container, the method comprising: clamping the container; puncturing a lid of the container by a puncturing device; capturing the lid by the puncturing device; rotating a blade and the container with respect to each other to cut the lid from the container; unclamping the container; and separating the puncturing device from the lid.
13. The method of claim 12, wherein motion of a handle in a first direction causes the clamping, the puncturing, and the rotating.
14. The method of claim 13, wherein motion of the handle in a second direction causes the unclamping, and the separating.
15. The method of claim 12, wherein the blade and the container rotate at least 120 degrees with respect to each other.
16. The method of claim 12, wherein the blade and the container rotate at least 360 degrees with respect to each other.
17. The method of claim 12, wherein the rotating is performed by a motor.
18. The method of claim 12, wherein the blade rotates and the container is stationary.
19. A container opener, comprising: a handle; a tab grabber to grab a tab on a lid of a container; a cutting blade; and an actuating mechanism operatively coupled to the handle, the tab grabber, and the cutting blade; wherein the actuating mechanism is configured to translate a rotational movement of the handle in a first direction to a movement of the tab grabber and the cutting blade in a vertical direction of the container opener such that the tab grabber grabs the tab of the lid of the container held in the container opener and the cutting blade cuts off the lid.
20. The container opener of claim 19, wherein the actuating mechanism is configured to translate a rotational movement of the handle in a second direction to separate the tab grabber from the lid.

21. A container opener, comprising: a handle; a cutting blade; and an actuating mechanism operatively coupled to the handle and the cutting blade; wherein the actuating mechanism is configured to translate a rotational movement of the handle in a first direction to a movement of the cutting blade in a vertical direction of the container opener and to rotational movement of the cutting blade such that the cutting blade cuts off a lid of a container.
